HOMEWORK 1 SOLUTIONS

SECTION 1.1

1 $\boxed{a}$

We substitute $x = 19t-35$, $y = 25-13t$, $z = t$ on each of the equations of the system:

- $2(19t-35)+3(25-13t)+t = 38t-70+75-39t+t = 5+0t = 5$ ✓
- $5(19t-35)+7(25-13t)-4t = 95t-175+175-91t-4t = 0$ ✓

So $x = 19t-35$, $y = 25-13t$, $z = t$ is indeed a solution for all values of t.

2 $\boxed{c}$

Solving for y: $y = 3x+2z-5$ (x, z free), giving us the solution

$$\boxed{\begin{aligned} x &= s \\ y &= 3s+2t-5 \quad , \quad s,t \text{ arbitrary} \\ z &= t \end{aligned}}$$

We could solve for z instead, to obtain

$$\boxed{\begin{aligned} x &= p \\ y &= q \quad , \quad p,q \text{ arbitrary} \\ z &= \tfrac{5}{2}-\tfrac{3}{2}p+\tfrac{1}{2}q \end{aligned}}$$

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$\boxed{a}$ $\left[\begin{array}{cc|c} 1 & -3 & 5 \\ 2 & 1 & 1 \end{array}\right]$

$\boxed{b}$ $\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 1 & 1 \end{array}\right]$

$\boxed{c}$ $\left[\begin{array}{ccc|c} 1 & -1 & 1 & 2 \\ 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & 0 \end{array}\right]$

$\boxed{d}$ $\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & 2 \end{array}\right]$

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$\boxed{a}$ $\begin{cases} x - y + 6z = 0 \\ \quad\; y \qquad\;\; = 3 \\ 2x - y \qquad = 1 \end{cases}$

$\boxed{b}$ $\begin{cases} 2x - y \qquad\quad = -1 \\ -3x + 2y + z = 0 \\ \qquad\;\; y + z = 3 \end{cases}$

9 $\boxed{d}$

$$\left[\begin{array}{cc|c} 3 & 4 & 1 \\ 4 & 5 & -3 \end{array}\right] \xrightarrow[R_1 \leftrightarrow R_2]{} \left[\begin{array}{cc|c} 4 & 5 & -3 \\ 3 & 4 & 1 \end{array}\right] \xrightarrow[R_1 - R_2]{} \left[\begin{array}{cc|c} 1 & 1 & -4 \\ 3 & 4 & 1 \end{array}\right] \xrightarrow[R_2 - 3R_1]{} \left[\begin{array}{cc|c} 1 & 1 & -4 \\ 0 & 1 & 13 \end{array}\right] \xrightarrow[R_1 - R_2]{} \left[\begin{array}{cc|c} 1 & 0 & -17 \\ 0 & 1 & 13 \end{array}\right]$$

Arguing as with the previous system, we conclude that this system has a unique solution:

$$\boxed{x = -17,\ y = 13}$$

14 $\boxed{a}$

<u>FALSE</u> : The system $\{x + y = 2$ has $n = 2$ variables and $m = 1$ equation, but the augmented matrix $[1\ 1 | 2]$ only has one row.

$\boxed{b}$

<u>FALSE</u> : The system from problem 10a is consistent but does not have infinitely many solutions.

$\boxed{c}$

<u>TRUE</u> : Row operations preserve solutions, so if we apply a row operation to a consistent system, the result is also consistent.

$\boxed{d}$

<u>TRUE</u> : Because row operations are reversible and preserve the solutions of a system, then the original system has to be inconsistent if it is carried to an inconsistent system via row operations.

SECTION 1.2.

2 $\boxed{a}$

$$\begin{bmatrix} 0 & -1 & 2 & 1 & 2 & 1 & -1 \\ 0 & 1 & -2 & 2 & 7 & 2 & 4 \\ 0 & -2 & 4 & 3 & 7 & 1 & 0 \\ 0 & 3 & -6 & 1 & 6 & 4 & 1 \end{bmatrix} \xrightarrow[-R_1]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & -1 & 1 \\ 0 & 1 & -2 & 2 & 7 & 2 & 4 \\ 0 & -2 & 4 & 3 & 7 & 1 & 0 \\ 0 & 3 & -6 & 1 & 6 & 4 & 1 \end{bmatrix} \xrightarrow[\substack{R_2 - R_1 \\ R_3 + 2R_1 \\ R_4 - 3R_1}]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 3 & 9 & 3 & 3 \\ 0 & 0 & 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 4 & 12 & 7 & -2 \end{bmatrix}$$

$$\xrightarrow[R_2 \leftrightarrow R_3]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 3 & 9 & 3 & 3 \\ 0 & 0 & 0 & 4 & 12 & 7 & -2 \end{bmatrix} \xrightarrow[\substack{R_3 - 3R_2 \\ R_4 - 4R_2}]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 6 & -3 \\ 0 & 0 & 0 & 0 & 0 & 11 & -10 \end{bmatrix} \xrightarrow[R_4 - 2R_3]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 6 & -3 \\ 0 & 0 & 0 & 0 & 0 & -1 & -4 \end{bmatrix}$$

$$\xrightarrow[R_3 \leftrightarrow R_4]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & -1 & -4 \\ 0 & 0 & 0 & 0 & 0 & 6 & -3 \end{bmatrix} \xrightarrow[-R_3]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 & 6 & -3 \end{bmatrix} \xrightarrow[R_4 - 6R_3]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & -27 \end{bmatrix}$$

$$\xrightarrow[-\frac{1}{27}R_4]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[\substack{R_3 - 4R_4 \\ R_2 - 2R_4 \\ R_1 - R_4}]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & -1 & 0 \\ 0 & 0 & 0 & 1 & 3 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[\substack{R_1 + R_3 \\ R_2 + R_3}]{} \begin{bmatrix} 0 & 1 & -2 & -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_1+R_2} \boxed{\begin{bmatrix} 0 & 1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}$$

The rank of this matrix is 4.

3 b

$$\begin{cases} x_1 - 2x_2 + 2x_4 + x_6 = 1 \\ x_3 + 5x_4 - 3x_6 = -1 \\ x_5 + 6x_6 = 1 \end{cases}$$

$x_2 = r$ $\quad x_4 = s$ $\quad x_6 = t$

Leading variables

$$\boxed{\begin{aligned} &x_1 = 1 + 2r - 2s + t, \quad x_2 = r, \quad x_3 = -1 - 5s + 3t \\ &x_4 = s, \quad x_5 = 1 - 6t, \quad x_6 = t, \quad r, s, t \text{ arbitrary} \end{aligned}}$$

c

$$\begin{bmatrix} 1 & 2 & 1 & 3 & 1 & 1 \\ 0 & 1 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1-3R_3} \begin{bmatrix} 1 & 2 & 1 & 0 & 4 & 1 \\ 0 & 1 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1-2R_2} \begin{bmatrix} 1 & 0 & 3 & 0 & 2 & -1 \\ 0 & 1 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{cases} x_1 + 3x_3 + 2x_5 = -1 \\ x_2 - x_3 + x_5 = 1 \\ x_4 - x_5 = 0 \end{cases}$$

$x_3 = s$ $\quad x_5 = t$

Leading variables

$$\boxed{\begin{aligned} &x_1 = -1 - 3s - 2t, \quad x_2 = 1 + s - t, \quad x_3 = s \\ &x_4 = t, \quad x_5 = t, \quad s, t \text{ arbitrary} \end{aligned}}$$

ALTERNATIVE WAY: *Use back-substitution*

$$\begin{cases} x_1 + 2x_2 + x_3 + 3x_4 + x_5 = 1 \\ x_2 - x_3 + x_5 = 1 \\ x_4 - x_5 = 0 \end{cases} \Rightarrow$$

$x_5 = t$, $x_4 = x_5 = t$, $x_3 = s$, $x_2 = 1 + x_3 - x_5 = 1 + s - t$,

$x_1 = 1 - 2x_2 - x_3 - 3x_4 - x_5 = 1 - 2(1+s-t) - s - 3t - t = -1 - 3s - 2t$

$x_3 = s$ $\quad x_5 = t$

Leading variables

$$\boxed{\begin{aligned} &x_1 = -1 - 3s - 2t, \quad x_2 = 1 + s - t, \quad x_3 = s \\ &x_4 = t, \quad x_5 = t, \quad s, t \text{ arbitrary} \end{aligned}}$$

5 d

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 2 & 5 & -3 & 1 \\ 1 & 4 & -3 & 3 \end{array}\right] \xrightarrow[R_3-R_1]{R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 2 & -2 & 1 \end{array}\right] \xrightarrow[R_3-2R_2]{} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 7 \end{array}\right]$$

Inconsistent, because it contain the equation $0=7$.

h

$$\left[\begin{array}{ccc|c} 1 & 2 & -4 & 10 \\ 2 & -1 & 2 & 5 \\ 1 & 1 & -2 & 7 \end{array}\right] \xrightarrow[R_3-R_1]{R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -4 & 10 \\ 0 & -5 & 10 & -15 \\ 0 & -1 & 2 & -3 \end{array}\right] \xrightarrow[-\frac{1}{5}R_2]{} \left[\begin{array}{ccc|c} 1 & 2 & -4 & 10 \\ 0 & 1 & -2 & 3 \\ 0 & -1 & 2 & -3 \end{array}\right] \xrightarrow[R_1-2R_2]{R_3-R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{cases} x = 4 \\ y - 2z = 3 \end{cases}$$

Leading $z=t$

$x=4$

$y=3+2t$ t arbitrary

$z=t$

9 c

$$\left[\begin{array}{ccc|c} -1 & 3 & 2 & -8 \\ 1 & 0 & 1 & 2 \\ 3 & 3 & a & b \end{array}\right] \xrightarrow[R_1\leftrightarrow R_2]{} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ -1 & 3 & 2 & -8 \\ 3 & 3 & a & b \end{array}\right] \xrightarrow[R_3-3R_1]{R_2+R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 3 & 3 & -6 \\ 0 & 3 & a-3 & b-6 \end{array}\right] \xrightarrow[\frac{1}{3}R_2]{} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -2 \\ 0 & 3 & a-3 & b-6 \end{array}\right] \xrightarrow[R_3-3R_2]{} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & a-6 & b \end{array}\right]$$

The system is inconsistent if and only if the last row is $[0\ 0\ 0 | b]$ and $b \neq 0$. This happens only if $a-6=0$ (i.e. $a=6$) and $b\neq 0$. Thus, the system is consistent if and only if $a\neq 6$ or $b=0$.

Case I : $a \neq 6$.

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & a-6 & b \end{array}\right] \xrightarrow[\frac{1}{a-6}R_3]{} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & \frac{b}{a-6} \end{array}\right]$$

- Rank of augmented matrix $=3=$ n° of variables

$\Rightarrow$ The system has a unique solution

Case II : $a=6$ (so $b=0$ in order for the system to be consistent)

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & a-6 & b \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

- Rank of augmented matrix $=2<3=$ n° of variables

$\Rightarrow$ Infinitely many solutions.

CONCLUSION:

- No solutions precisely when $a=6$ and $b=0$
- Unique solution precisely when $a \neq 6$
- Infinitely many solutions precisely when $a=6$ and $b \neq 0$